

Journal Pre-proof

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PII: S0021-8502(21)00540-1

DOI: <https://doi.org/10.1016/j.jaerosci.2021.105809>

Reference: AS 105809

To appear in: *Journal of Aerosol Science*

Received Date: 30 January 2021

Revised Date: 1 May 2021

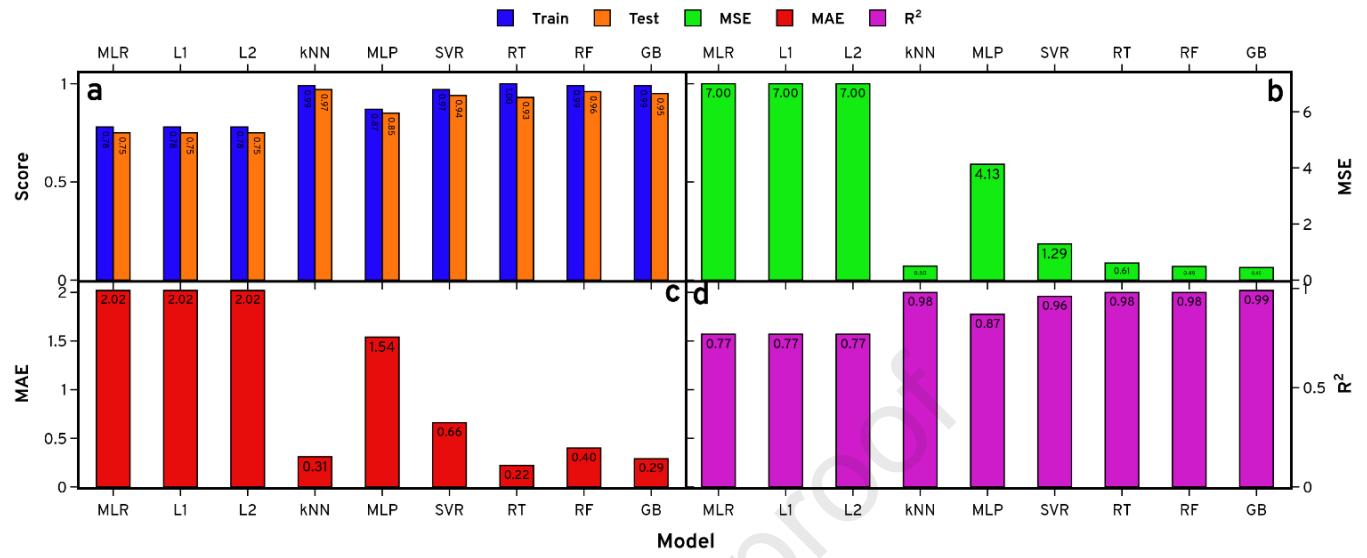
Accepted Date: 4 May 2021

Please cite this article as: Kumar V, Sahu M, Evaluation of nine machine learning regression algorithms for calibration of low-cost PM_{2.5} sensor, *Journal of Aerosol Science* (2021), doi: <https://doi.org/10.1016/j.jaerosci.2021.105809>.

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Graphical Abstract



Evaluation of Nine Machine Learning Regression Algorithms for Calibration of Low-Cost PM_{2.5} Sensor

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Submitted to:
Journal of Aerosol Science

January 31, 2021

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Abstract

Low-cost sensors (LCS) can construct a high spatial and temporal resolution $PM_{2.5}$ network but are affected by environmental parameters such as relative humidity and temperature. The data generated by LCS are inaccurate and require calibration against a reference instrument. This study has applied nine machine learning (ML) regression algorithms for Plantower PMS 5003 LCS calibration and compared their performance. The nine ML algorithms applied in this study are: (a) Multiple Linear Regression (MLR); (b) Lasso regression (L1); (c) Ridge regression (L2); (d) Support Vector Regression (SVR); (e) k- Nearest Neighbour (kNN); (f) Multilayer Perceptron (MLP); (g) Regression Tree (RT); (h) Random Forest (RF); (i) Gradient Boosting (GB). The comparison exhibits that kNN, RF and GB have the best performance out of all the algorithms with train scores of 0.99 and test scores of 0.97, 0.96 and 0.95 respectively. This study validates the capability of ML algorithms for the calibration of LCS.

Keywords: $PM_{2.5}$, low-cost sensor, machine learning, calibration, random forest, gradient boosting

23

24 **1. Introduction**

25 $PM_{2.5}$ exposure affects more humans globally than any other air pollutant and caused 4.2
 26 million deaths, i.e., 7.6% of global deaths in 2015 and is the 5th leading risk factor for death
 27 (Forouzanfar et al., 2016; Cohen et al., 2017; WHO, 2018). $PM_{2.5}$ measurement is significant
 28 for the formulation of air pollution control measures, policies and frameworks to counter the
 29 potentially damaging effects on humans and climate change. For example, the National Clean
 30 Air Program (NCAP) launched by GoI to monitor and control PM emissions in Indian cities
 31 (Ganguly et al., 2020). There are two primary methods to measure $PM_{2.5}$ viz. Federal
 32 Reference Method (FRM) and Federal Equivalent Method (FEM) (Noble et al., 2001). FRM
 33 requires a gravimetric method where particle mass concentration is determined by weighing
 34 the filters before and after the sampling period as a measurement technique. FRM is the most
 35 accurate method and is widely used by regulatory bodies. However, certain disadvantages of
 36 the FRM method, such as it is not a real-time form of measurement, i.e., it only provides a
 37 24-h average. FRM method also has a high operation cost and has manual process involved
 38 and is not portable (Ayers et al., 1999; Noble et al., 2001; Le et al., 2020). Beta attenuation
 39 monitor (BAM) and tapered element oscillating microbalance (TEOM) are two standard
 40 FEM methods. FEM methods have a higher temporal resolution, i.e., provide 1-h average and
 41 have relatively low operational cost compared to FRM but have high installation cost (Ayers
 42 et al., 1999; Chung et al., 2001; Le et al., 2020).

43 $PM_{2.5}$ concentration is mostly organized spatiotemporally and thereof; the concentration and
 44 exposure can vary notably even in the nearby areas. The FRM and FEM cannot be installed
 45 in large numbers to create a dense network of high spatial resolution due to the lack of
 46 portability and high installation as well as operational cost. Low-cost sensors (LCS) can be a

possible alternative and can be installed in large numbers to construct a high spatial and temporal resolution $PM_{2.5}$ network (Di Antonio et al., 2018; Bulot et al., 2020). LCS are cost-effective compared to FRM/FEM and can collect real-time data, i.e., minute resolution. LCS are compact, light-weight, portable and require low maintenance. However, there are certain shortcomings associated with using LCS which need to be addressed before being used extensively. LCS are based on laser light scattering (LLS) technology and uses a light beam to estimate the concentration based on the light scattered by particles passing through an air stream (Badura et al., 2018) instead of measuring their mass concentration or counting particles directly (He et al., 2020). Aerosol composition and size distributions, meteorological conditions such as relative humidity (RH) and temperature (T) can significantly influence the performance and accuracy of LCS (Rai et al., 2017; Badura et al., 2018; Zheng et al., 2018; Loh and Choi, 2019; Bai et al., 2020; Chu et al., 2020; Qin et al., 2020; Zusman et al., 2020). In high RH conditions, the aerosol particles absorb water and change the size, morphology, refractive index etc., known as hygroscopic growth (Lee et al., 2008), which regulates the amount of light scattered and so the particle concentration. The critical reason for RH and T's inclusion while calibrating the LCS with an FRM/FEM is the difference in operational RH and T of these instruments during measurement. FRM/FEM instruments generally operate at normal conditions (T: 20-23 °C, RH: 30-40%), while the LCS measures data in ambient conditions, which can cause disagreement in reported concentrations (Malings et al., 2020). The $PM_{2.5}$ concentration from the LCS alone is not adequate to explain the disparity in the LCS measurements (Zheng et al., 2018) and so RH and T are the crucial parameters for the LCS calibration (Lee et al., 2020). Also, the introduction of RH and T in the calibration model significantly improves the accuracy and can account for up to 17% and 7% variation in the $PM_{2.5}$ measurement reported by the low-cost sensor respectively which aids in achieving

the highest accuracy possible (Gao et al., 2015; Rai et al., 2017; Chen et al., 2018; Lin et al., 2018; Zheng et al., 2018; Lee et al., 2020; Jiang et al., 2021).

Therefore, even though the cost of LCS is an advantage compared to FRM/FEM, the accuracy of concentration reported by the LCS is questionable (Li et al., 2020) and can vary concerning site/season due to change in environmental conditions. So, the LCS requires rigorous calibration against a reference instrument to account for RH and T's effect to meet regulatory instruments standards. Although LCS manufacturers develop calibration factors for the sensors but in laboratory conditions, which is not the same as the site where the sensor is installed and so the calibration factors become incompetent. So, it is vital to re-calibrate the sensors at the installation site to report accurate concentrations (Badura et al., 2018; Di Antonio et al., 2018; Bulot et al., 2020; Si et al., 2020). Besides, LCS may also require developing multiple calibration models specific to site or season and regular calibration to ensure accuracy of the particle concentration reported (Miskell et al., 2018; Zheng et al., 2018). Due to this, calibration of LCS manually can be hectic and is not a feasible option and some sophisticated models need to be developed. The calibration models have to be highly accurate and efficient in terms of time and computational power with the ability to handle large datasets (Wang et al., 2020). Machine learning can play a crucial role in this due to the wide range of algorithms for various kinds of datasets and applications.

So far, many statistical techniques such as Gaussian Process regression (GPR) and simple/multiple linear regression (Badura et al., 2019; Zheng et al., 2019; Si et al., 2020; Patra et al., 2021) have been implemented for calibrating the LCS with the reference instruments. A limited number of studies have also applied machine learning (ML)

algorithms such as k-nearest neighbors (Loh and Choi, 2019), support vector machines (Loh and Choi, 2019; Wang et al., 2020), artificial neural network (Badura et al., 2019; Si et al., 2020), decision tree (Wijeratne et al., 2019), random forest (Loh and Choi, 2019; Wang et al., 2019; Wang et al., 2020) and gradient boosting (Johnson et al., 2018; Loh and Choi, 2019; Si et al., 2020). The problem with ML models is that they can suffer from overfitting. Overfitting is when the model learns the training data too well and provides a high train score but lower test score. Overfitted models are unable to perform accurately on new datasets and cannot be generalized. One way to overcome overfitting is to train multiple ML algorithms of a wide range and working principles and evaluate their performance to find the most appropriate model for the problem. No attempts have been made previously to apply a range of ML algorithms for LCS calibration.

The following objectives are addressed in this paper: (a) to investigate the performance of Plantower PMS 5003 LCS compared to Thermal Fisher Scientific SHARP model 5030 as a reference method for measurement of $PM_{2.5}$ at Alberta, Canada (Si, 2019) and (b) evaluate and compare the performance of nine ML algorithms for calibration of LCS. The nine ML algorithms applied in this study are: (a) multiple linear regression (MLR); (b) Lasso regression (L1); (c) Ridge regression (L2); (d) support vector regression (SVR); (e) k -nearest neighbour (kNN); (f) multilayer perceptron (MLP); (g) regression tree (RT); (h) random forest (RF); (i) gradient boosting (GB). The predicted values from all the algorithms are compared with the LCS and reference instrument measurements and each other.

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123 **2. Methodology**

124 **2.1. Data**

125 The Plantower PMS 5003 LCS is evaluated in this study and measures $PM_{2.5}$, RH and T. The
 126 Thermal Fisher Scientific SHARP model 5030 is used as a reference instrument to evaluate
 127 the LCS. The data used in this study (Si, 2019) was collected at Calgary Region Airshed
 128 Zone (CRAZ) in Calgary, Alberta, Canada from 7 December 2018 to 26 April 2019 at an
 129 interval of 6s. The data was aggregated to derive the hourly concentrations. The exact
 130 sampling method, details of instruments, and the data pre-processing techniques applied to
 131 the data can be found in Si et al. (2020).

132

133 **2.2. Algorithms**

134 Regression is a powerful and fundamental statistical technique in machine learning and finds
 135 its application in economics, psychology, geography, and so forth (Sammut and Webb, 2011;
 136 Mendenhall and Sincich, 2014). Regression analysis is the study of dependence or relation
 137 between random variables of interest and infers mathematical functions to explain their
 138 behaviour, known as regression models (Rawlings et al., 1998; Mendenhall and Sincich,
 139 2014). Regression analysis requires two types of real-valued variables, target/dependent and
 140 independent represented by y and x respectively. The objective of regression is to map a
 141 function such that $y = f(x) + \varepsilon$, where ε is the error (Kroese et al., 2019; Zaki and Meira,

2020). The cases in which there is more than one independent variable are known as multi-regression, where the equation transforms to $y = f(x_1, x_2, x_3, \dots, x_n) + \varepsilon$, where $(x_1, x_2, x_3, \dots, x_n) \in x$. In this study, we have three independent variables viz. $PM_{2.5}$, temperature, and relative humidity from the LCS represented by $PM_{2.5_LCS}$, T , and RH respectively and one target variable, $PM_{2.5}$ from the reference instrument represented by $PM_{2.5_REF}$. The regression algorithms applied in this study attempt to estimate functions to explain the effect of temperature and relative humidity on the measurement of $PM_{2.5}$ from the LCS to calibrate it with the reference instrument measurements. Mathematically, the objective function is $PM_{2.5_REF} = f(PM_{2.5_LCS}, Temp, RH) + \varepsilon$. Nine different ML regression algorithms were applied in this study which is discussed briefly below.

2.2.1. Linear models

Linear regression (LR) is the simplest regression model as it involves only one independent variable and assumes a linear function (straight line) between x and y (Bishop, 2006; Mendenhall and Sincich, 2014). The objective of LR is to fit a line ($y = w \cdot x + b$, where b is the bias/intercept and w is the slope/weight) to a set of data points (x) to predict future values of y for given values of x , known as best fit line (Rawlings et al., 1998; Mendenhall and Sincich, 2014). The slope and intercept can be calculated using equations (1) and (2) respectively.

$$w = \frac{SS_{xy}}{SS_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n(\bar{x})^2} \dots (1)$$

$$b = \bar{y} - w \cdot \bar{x} \dots (2)$$

where $\bar{x}$ and $\bar{y}$ is the mean of x and y respectively and n is the sample size. The deviation between the actual (y_{act}) and predicted (y_{pred}) value of y is known as error. LR applies the

least square error criterion to find out the best fit line. The best fit line is the one where the sum of the errors (SE) is zero and the sum of the squares of the errors (SSE) is minimum (Mendenhall and Sincich, 2014). SSE is the cost function (CF) of LR. SE and SSE can be calculated using equations (3) and (4) respectively.

$$SE = \sum_{i=1}^n (y_{pred_i} - y_{act_i}) = \sum_{i=1}^n (w \cdot x_i + b - y_{act_i}) \quad \dots (3)$$

$$SSE = \sum_{i=1}^n (y_{pred_i} - y_{act_i})^2 = \sum_{i=1}^n (w \cdot x_i + b - y_{act_i})^2 \quad \dots (4)$$

Multiple linear regression (MLR) is an extended version of LR and involves more than one independent variable. In MLR, the objective is to find the best-fit plane instead of line and the equation broadens to the form of $y = b + w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n$, where $x_1, x_2, x_3, \dots, x_n$ are the independent variables and $(w_1, w_2, w_3, \dots, w_n) \in w$ are the respective weights and w is the weight vector (Mohri et al., 2018; Kroese et al., 2019; Zaki and Meira, 2020). The method of fitting for MLR is identical to LR.

For regression, the trained model's error can be fragmented as the sum of error due to variance and error due to bias, known as bias-variance decomposition. Generally, mean square error (MSE) is used to quantify the prediction error which is equivalent to the sum of variance and squared bias, as seen in equation 5. The bias component describes how the average predicted value differs from the mean of actual values. In contrast, the variance component explains predicted values' inconsistency for a given data point from different models (Bishop, 2006; Hastie et al., 2009; Sammut and Webb, 2011; Fortmann-Roe, 2012).

$$MSE(x) = E[(f(x) - y)^2] = (E[f(x)] - y)^2 + E[f(x) - E[f(x)]]^2 = \text{bias}^2 + \text{variance} \quad \dots (5)$$

A high bias suggests that the model does not fit the data well and incorporates more assumptions about the target variable whereas a low bias means the model made fewer assumptions. A high variance value reflects random noise in the training data while a low value suggests predicted values close to each other. A high bias value causes low variance in order to reduce the model error according to equation 5, which leads to a phenomenon known as underfitting. In underfitting, the model cannot capture the underlying trend between dependent and independent variables and neither performs well on the training nor the test set, producing high prediction error for both. Increasing the model complexity reduces training error but too much training increases variance on the cost of bias causing overfitting in which the model adapts to the data too well. Overfitted models do not generalize well on other datasets and produce low training but comparatively larger test errors. There is a constant trade-off between bias and variance to find the optimum value in order to get the least model error possible, known as bias-variance dilemma as illustrated in Figure 1 (Bishop, 2006; Hastie et al., 2009; Sammut and Webb, 2011; Fortmann-Roe, 2012).

Regularization is a technique to control overfitting which discourages the regression weights from reaching large values by decreasing the variance at the expense of increasing the bias slightly. Regularization adds a penalty term to the cost function depending on the weights' norm (Bishop, 2006; Hastie et al., 2009; Flach, 2012; Kroese et al., 2019). Two different regularization techniques are applied in this study Lasso (L1) and Ridge (L2) have subtle but significant differences. Lasso applies the L1 (Manhattan/Taxicab) norm while Ridge applies the L2 (Euclidean) norm. Due to this, L2 regression weights may remain small but still non-zero, while L1 favors sparse model and drives many weights to zero acting as a feature extraction method, especially in the case of multi-regression (Bishop, 2006; Hastie et al., 2009; Flach, 2012; Zaki and Meira, 2020). The regularized cost function's for L1 and L2 are in equation (6) and (7):

$$CF_{L1} = \sum_{i=1}^n (w \cdot x_i + b - y_{act_i})^2 + \lambda |w| \dots (6)$$

$$CF_{L2} = \sum_{i=1}^n (w \cdot x_i + b - y_{act_i})^2 + \lambda w^2 \dots (7)$$

where $\lambda (\geq 0)$ is the regularization constant. λ controls the trade-off between the regularization and SSE components of the regularized cost function. When $\lambda = 0$, there is no regularization and the model will have low bias and possibly high variance. However, if $\lambda \rightarrow \infty$, all the weights would tend to zero producing a low variance and high bias overfitted model. Varying the λ estimates the best balance between bias and variance for an optimal predictive model. A small positive value of λ always guarantees a solution (Bishop, 2006; Hastie et al., 2009; Sammut and Webb, 2011; Flach, 2012; Mohri et al., 2018). L1 and L2 are simply LR with a normed cost function and so the method of fitting remains the same. Further details about these techniques can be found in Rawlings et al. (1998), Bishop (2006), Hastie et al. (2009), Mendenhall and Sincich (2014).

Support Vector Regression (SVR) is a powerful and robust modeling technique in ML. SVR follows the same paradigm of finding a function that fits the training data well to minimize the prediction error as discussed in LR but with specific crucial differences. SVR attempts to construct a tube of width $\varepsilon > 0$ (user-specified) around the regression function and treats deviations outside the margin as noise. A small value of ε may lead to a tube that does not enclose the entire data, while a very high value would mean that outliers are the only points that define the regression equation and produce an insignificant prediction model. SVR tries to minimize the absolute error instead of SSE as in LR (Gunn, 1998; Bishop, 2006; Hastie et al., 2009; Mohri et al., 2018; Yang, 2019). SVR implements a loss function known as ε -insensitive loss defined as:

$$L_{\varepsilon}(y) = \begin{cases} 0 & \text{for } |w \cdot x + b - y| < \varepsilon \\ |w \cdot x + b - y| - \varepsilon & \text{otherwise} \end{cases} \dots (8)$$

Due to the ε -insensitive loss function, only the deviation of points outside the tube contributes to the final error. The deviation of points inside the tube is ignored in the optimization. The points on the border of the tube and outside are known as support vectors as they support the regression line. The ε -insensitive loss function reduces the risk of overfitting as large outliers have a restricted effect on the regression equation. Mathematically, the optimization problem for SVR can be written as:

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n |y_i - (w \cdot x_i + b)| \dots (9)$$

subject to the constraints:

$$\begin{cases} y_i - w \cdot x_i - b \leq \varepsilon \\ w \cdot x_i + b - y_i \leq \varepsilon \end{cases} \dots (10)$$

Where $C > 0$ is the penalty parameter that controls the trade-off between amount up to which deviation greater than ε are tolerated and flatness of the regression model (tube) (Gunn, 1998; Witten and Frank, 2005; Clarke et al., 2009; Kuhn and Johnson, 2013; Yang, 2019). More details on the derivation and implementation of SVR can be found in Gunn (1998), Clarke et al. (2009), Kuhn and Johnson (2013) and Yang (2019).

2.2.2. *k*-Nearest Neighbour Regression

k-Nearest Neighbour (kNN) is a simple and easy to implement learning-by-memorizing-based ML algorithm. kNN is developed on the assumption that similar things exist near each other and can be applied for regression and classification. kNN does not create any stringent abstractions about the data's underlying structure. Instead, it merely memorizes the training

data and predicts the value for new instances based on the closest samples. kNN algorithm is a five-step process: (a) select distance metric; (b) select number of nearest neighbours ($k < n$); (c) compute distance from other data points to desired point; (d) sort the points in increasing order of distance; (e) compute the average of k nearest neighbours' responses. Euclidean distance is the most commonly used metric for regression, although some other metrics are available (Hastie et al., 2009; Sammut and Webb, 2011; Kuhn and Johnson, 2013; Yang, 2019). For a p dimensional and n sample size data, the Euclidean distance between two points with attribute values $a_1, a_2, a_3, \dots, a_p$ and $b_1, b_2, b_3, \dots, b_p$ can be calculated using equation (11) and the value of response variable for new instance can be calculated using equation (12):

Euclidean Distance

$$= \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2 + \dots + (a_p - b_p)^2} \dots (11)$$

$$y_{new}(x_{new}) = \frac{1}{k} \sum_{i \in k(x_{new})} y_i \dots (12)$$

where $k(x_{new})$ is the set of k nearest neighbours of x_{new} . k decides the accuracy of the model and so choosing the optimal value of k is of critical significance. A minimal value of k pulls few data points for accurate estimation and becomes sensitive to noise in data accepting high variance in predictions that may lead to overfitting. While a very high value of k may include irrelevant data points that are not actual neighbours and increases the bias at the cost of variance, reducing the model's accuracy. kNN is also known as lazy algorithm as it is computationally expensive and significantly slower for large datasets since it needs to compute and sort the distances for the entire training dataset every time to find the nearest neighbours. This makes kNN inappropriate for scenarios where rapid predictions are required (Hastie et al., 2009; Sammut and Webb, 2011; Kuhn and Johnson, 2013; Yang, 2019).

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266 **2.2.3. Multilayer Perceptron**

267 Multilayer Perceptron (MLP) is a feed-forward (connections between nodes do no form loop)
 268 artificial neural network (ANN). The primary computing device of MLP is perceptron, a
 269 mathematical model of a neuron. MLP is a network of connected perceptrons organized in a
 270 layered formation. Each perceptron possesses an activation function and weights for each
 271 input. The objective of MLP is to learn these weights from the data such that the prediction
 272 accuracy is optimized. MLP consists of one input and output layer and can have more than
 273 one hidden layer.

274 The working mechanism of MLP is straightforward and can be summarized as a five-step
 275 process: (a) initialize the input data and random but small weights; (b) train the network in
 276 the forward direction and predict the output; (c) calculate the cost/error (MSE is used mainly
 277 for regression); (d) backpropagate the error and update the weights accordingly; (e) repeat the
 278 steps (b) and (c) until the error is marginal (Shalev-Shwartz and Ben-David, 2014; Kubat,
 279 2017; Kroese et al., 2019; Yang, 2019; Zaki and Meira, 2020). MLP's are effective modeling
 280 techniques and can deal with several kinds of data. MLP falls into deep learning, a set of ML
 281 which deals with neural networks and is extensive. The mathematical derivations associated
 282 with training of the network and backpropagation of error are beyond the scope of this paper
 283 and can be found in Shalev-Shwartz and Ben-David (2014), Kubat (2017) and Yang (2019).

284

285 **2.2.4. Tree-based models**

286 Tree-based models are among the most popular ML models due to their iterative divide-and-
 287 conquer nature and have its root in the data structure. They are easy to implement and
 288 efficient but are computationally intensive. The tree-based model constructs a set of highly

interpretable logical (if-then) conditions by recursively partitioning the decision space into smaller subspaces using training data and presents the decision process in the form of a tree graphically. They implicitly perform feature selection and can be applied for both regression and classification on datasets with large numbers of cases and/or variables (Hastie et al., 2009; Sammut and Webb, 2011; Flach, 2012; Yang, 2019).

There are numerous techniques for constructing a tree. However, here we follow the procedure of constructing the regression tree (RT) from one of the most utilized frameworks called classification and regression trees (CART) of Breiman et al. (1984). For a regression problem, the entire training dataset (D) is initially at the tree's root node and specific logical tests are conducted. The test partitions the data into two group (D_1 and D_2), one with the values satisfying the test and remaining to the other such that the overall SSE is minimized:

$$SSE = \sum_{i \in D_1} (y_i - \bar{y}_{D_1})^2 + \sum_{j \in D_2} (y_j - \bar{y}_{D_2})^2 \dots (13)$$

where $\bar{y}_{D_1}$ and $\bar{y}_{D_2}$ are the mean of the training set predictions for D_1 and D_2 respectively. The process is repeated until convergence criteria are met and no more splits are possible. The last nodes of the tree are called decision nodes (or leaves) and contain the predicted value for the target variable (Bishop, 2006; Hastie et al., 2009; Sammut and Webb, 2011; Kuhn and Johnson, 2013; Yang, 2019).

Bagging (short for bootstrap aggregating) and boosting are two popular ensemble learning techniques (models that combine the output of multiple models) in ML. The main idea behind bagging and boosting is constructing multiple decision trees and combining their predictions by averaging (in regression) or voting (in classification) to reduce variance and increase prediction accuracy. The difference is that bagging creates individual trees and assigns equal weight to all trees. Contrary to that, in boosting the new trees are influenced by the previous ones' performance and assigns weight based on the trees' performance (Sutton, 2005; Bishop,

2006; Hastie et al., 2009; Yang, 2019). Two ensemble techniques used in this study viz. Random Forest (RF) and Gradient Boosting (BF) are extensions of bagging and boosting respectively. The detailed discussion of these algorithms is beyond the scope of this paper but can be seen in Sutton (2005), Hastie et al. (2009) and Yang, (2019).

3. Results and Discussion

3.1 Evaluation of LCS measurements

LCS measurements under ambient conditions are affected by high relative humidity (RH) and sensitive to temperature (T). The comparative analysis of $PM_{2.5}$ measurements of LCS ($PM_{2.5_LCS}$) and reference instrument ($PM_{2.5_REF}$) indicates a high variation in measured values, sometimes almost double when the value is above $10 \mu g/m^3$. For values less than that, the values are closer comparatively. The mean($\pm$ sd), minimum and maximum values measured by reference instrument are 6.57($\pm$ 5.57), 0.00 and 38.87 respectively, while for LCS the values are 9.89($\pm$ 12.09), 0.03 and 77.8. The statistical performance parameters between $PM_{2.5_REF}$ and $PM_{2.5_LCS}$ are calculated and presented in table 1.

This drift produced in the LCS measurements is due to the variation in RH or T and sometimes a combination of both, as shown in Figure 2. The reference instrument's measurements are not affected by RH and T due to a drying system. Some other environmental parameters might also be affecting the LCS measurements but we are ignoring them in this study due to lack of data. The objective is to calibrate the $PM_{2.5_LCS}$ with the $PM_{2.5_REF}$ that incorporates RH and T's effect and reduce the measurements' variance. In this study, we have applied nine ML regression algorithms to calibrate the LCS.

3.2 Calibration results

To understand the regression algorithms' capabilities applied in this study, the trained models need to be tested on new data (unavailable to the model) to check the model's performance. The entire dataset was split into two parts: train and test (70/30). The entire pipeline is divided into two steps: parameter optimization and training/prediction. In parameter optimization, a grid search was performed to infer appropriate parameters for the data with ten cross-fold validation for all the models simultaneously. The second step comprises applying the parameters derived from parameter optimization and results produced by the models discussed below.

3.2.1 Linear Models

L1 and L2 models are producing the same result as MLR. A regularization constant (λ) of 0.01 and 0.1 was applied in L1 and L2 models respectively inferred through grid search. SVR model was trained on $\epsilon=0.2$ and penalty parameter (C) = 1. Out of four linear models, MLR, L1 and L2 have a low train (0.78) and test score (0.75) and only SVR is producing a high train (0.97) and test score (0.94). The linear models except for SVR, i.e., MLR/L1/L2, have reduced the mean (6.63 ± 5.03) and maximum (35.62) values of LCS measurements close to the range of reference instrument measurements but are unable to provide an accurate prediction as can be seen in Figure 3(a). Along with the train and test scores being low for these models, the MSE (7.00) and MAE (2.02) is low compared to actual values but is high compared to other models and vice-versa for R^2 (0.77). The weights of $PM_{2.5_LCS}$, RH and T are 0.46, -0.11 and -0.03 respectively, and the bias is 7.94.

This is a case of under-fitting where the model cannot completely identify the underlying relation between independent and dependent variables distribution. The regularization techniques also cannot handle the complexity of the problem and do not show any improvement compared to MLR. On the other hand, SVR exhibits high train and test scores and a higher R^2 between actual and predicted values than its siblings. But higher MSE (1.29) and MAE (0.66) sometimes almost double compared to the values of other algorithms such as kNN (0.31) and GB (0.29). SVR can also fit the line to the data without getting sensitive to outliers, as illustrated in Figure 3(b).

3.2.2 Multilayer Perceptron

MLP model's train (0.87) and test (0.85) scores are higher than MLR, L1 and L2 but lower than SVR. The MLP was trained with three hidden layers of size five each and tanh as activation function and some other peripheral parameters. The MSE (4.13) and MAE (1.54) between the model's predicted and actual values are relatively high compared to some models. The R^2 (0.87) value is significantly low compared to other algorithms. The predicted data is still susceptible to outliers, as shown in the scatter plot in Figure 4(a). MLP can be extremely powerful if trained regressively and more accurate models could be produced. We acknowledge that the model could not be trained with more hidden and deep layers due to computational competency.

3.2.3 k- Nearest Neighbour

kNN is building a very accurate model with train and test scores as high as 0.99 and 0.97. The kNN model was trained for $k=2$. The model's accuracy decreases on increasing the value of k . The MSE, MAE and R^2 values for kNN are 0.5, 0.31 and 0.98, respectively. The model's

prediction and actual values can be fitted by a line that explains their relationship without getting affected by outliers/noise in data, as illustrated in Figure 4(b).

3.2.4 Tree models

All three tree-based models produce quality results with train scores of 1.00, 0.99 and 0.99 and test scores of 0.93, 0.96, and 0.95 for RT, RF and GB, respectively. RT, RF and GB are trained with trees of maximum depth 28, 15 and 5 respectively. RT is producing the best train score. However, the difference between train and test scores for RT (0.07) is higher compared to RF (0.03) and GB (0.04). RT is capable of overfitting as they apply greedy algorithms due to which sometimes the optimal tree cannot be found. This is a case of overfitting. The RT model is learning the training data too well and cannot be generalized. RF and GB are capable of fixing the overfitting issue of RT, as seen in this case. The scatter plots for RT, RF and GB can be seen in Figures 5(a), (b) and (c) respectively. The data points for RF and GB are closer to the fit line compared to RT. The MSE (0.61) for RT is higher compared to RF (0.49) and GB (0.45) but MAE (0.22) is lower than RF (0.4) and GB (0.29). The R^2 value is in the same range. RF and GB are providing the best results on all parameters.

A comparison of the train and test scores (R^2) for the models is presented in Figure 6(a). The MSE, MAE and R^2 between actual ($PM_{2.5_REF}$) and values predicted from the models were compared and are presented in Figures 6(b), (c), and (d) respectively. The MSE is a measurement of goodness of fit. Lower MSE means a lower error or better fit. The overall evaluation and comparison of all the nine regression algorithms exhibit that kNN, RF and GB are the best-suited LCS calibration models.

3.2.5 Comparison of computation time

Another predominant parameter that dictates the real-time application of a calibration methodology is the time taken to train the calibration model. Since the calibration methods applied in this study are in two stages, the computation time is also segregated accordingly and is presented in Figure 7. The less complex computational models have their parameter optimization time of less than 10 minutes. For more complex high-end ML algorithms, the parameter optimization time is around 20 minutes, with SVR (22.35 minutes) taking the most time. The training time is closer to 5 s for almost all the algorithms except SVR taking 8.63 s. The prediction time is less than 2.5 s for all the algorithms. The evaluation of computation time displays that none of the high-end ML algorithms that produce accurate results have an advantage concerning computational time for this dataset. However, the computational time can depend based on the size of the dataset used and the computational power available to the modeler.

3.2.6 Comparison with previous studies

The results obtained on applying the algorithms are compared with previous studies, which have applied the respective algorithms and are presented in Table 2. It is clear from the comparison that the less complex computation models provide better results than Si et al. (2020) but are still underfitting. On the contrary, the high-end ML algorithms produce highly accurate results in this study, similar to the previous studies. Different techniques for hyperparameter optimization and differences in train test split ratio could be the reason for difference in the performance of the algorithms compared to Si et al. (2020). The primary criteria that affect the algorithms' performance are correct parameters for the respective algorithms and the selection of essential variables to achieve the best results. We

acknowledge that this study applied the algorithms on a particular dataset measured at a specific location. It would be interesting to investigate these algorithms' performance for multi-location and seasonal datasets and compare their performance, which will be conducted in future studies.

4. Conclusion

Nine machine learning (ML) regression algorithms have been examined to calibrate low-cost sensors (LCS). This was demonstrated on the dataset collected from a Plantower PMS 5003 LCS and Thermal Fisher Scientific SHARP model 5030 as a reference instrument. Multiple Linear (MLR), Lasso (L1) and Ridge (L2) regression models giving the same result and are underfitting the data while the Regression Tree (RT) is overfitting. Support Vector Regression (SVR) and Multilayer Perceptron (MLP) are giving good results compared to MLR, L1 and L2 but can be improved with increased computational capability. K- Nearest Neighbour (kNN), Random Forest (RF) and Gradient Boosting (GB) provide highly accurate results. kNN, RF and GB have the best performance out of all the algorithms with train scores of 0.99 and test scores of 0.97, 0.96 and 0.95 respectively. The overall evaluation and comparison demonstrate that kNN, RF and GB are the best-suited models for the calibration of LCS.

Data Availability. The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.3473833> (Si, 2019).

453

454 **Acknowledgments:**

455 This work has been supported by the Central Pollution Control Board as a part of the study
 456 “Pilot Study for Assessment of Reducing Particulate Air Pollution in Urban Areas by Using
 457 Air Cleaning System (sometimes called as Smog Tower)” (Grant no: RD/0120-CPCB000-
 458 001). Partial support from the study “Application of Nanoparticles in ESP for Inactivation of
 459 Microorganisms and Degradation of VOCs for Air Purification” (Grant no: RD/0119-
 460 DST0000-048) is acknowledged.

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Table Captions

Table 1. Statistical performance parameters between $PM_{2.5_REF}$ and $PM_{2.5_LCS}$

Table 2. Comparison with previous studies

Figure Captions

Figure 1. Bias-variance dilemma (Hastie et al., 2009; Fortmann-Roe, 2012)

Figure 2. Effect of relative humidity and temperature on LCS $PM_{2.5}$ measurements

Figure 3. Scatter plot between $PM_{2.5_REF}$ and values predicted from (a) MLR/L1/L2, (b) SVR

Figure 4. Scatter plot between $PM_{2.5_REF}$ and values predicted from (a) MLP, (b) kNN

Figure 5. Scatter plot between $PM_{2.5_REF}$ and values predicted from (a) RT, (b) RF, (c) GB

Figure 6. Comparison of (a) train and test scores, (b) MSE, (c) MAE, (d) R^2 for the nine ML regression algorithms

Figure 7. Comparison of computation time for the algorithms

Table 1. Statistical performance parameters between $PM_{2.5_REF}$ and $PM_{2.5_LCS}$

Parameter	$PM_{2.5_REF}$ vs $PM_{2.5_LCS}$
R^2	0.74
MSE	72.24
MAE	4.95
Slope	1.86 ± 0.02
Intercept	-2.38 ± 0.17

Table 2. Comparison with previous studies

Model	References	R^2	MSE	MAE
MLR	Johnson et al., 2018	0.44	10.98	-
	Badura et al., 2019	0.85	19.68	-
	Si et al., 2020	0.6	21.61	3.09
	Wang et al., 2020	0.76	47.33	-
	This study	0.77	7	2.02
L2	Johnson et al., 2018	0.45	10.78	-
	This study	0.77	7	2.02

SVR	Loh and Choi, 2019	0.76	31.36	-
	Wang et al., 2020	0.94	27.3	-
	This study	0.96	1.29	0.66
kNN	Loh and Choi, 2019	0.78	29.16	-
	This study	0.98	0.5	0.31
	Badura et al., 2019	0.79	26.62	-
MLP/ANN	Si et al., 2020	0.67	17.61	2.63
	This study	0.87	4.13	1.54
RT	Wijeratne et al., 2019	0.99	-	-
	This study	0.98	0.61	0.22
	Loh and Choi, 2019	0.8	27.04	-
RF	Wang et al., 2019	0.98	-	-
	Wang et al., 2020	0.94	10.34	-
	This study	0.98	0.49	0.4
	Johnson et al., 2018	0.72	5.51	-
GB	Loh and Choi, 2019	0.82	24.5	-
	Si et al., 2020	0.72	15.26	2.38
	This study	0.99	0.45	0.29

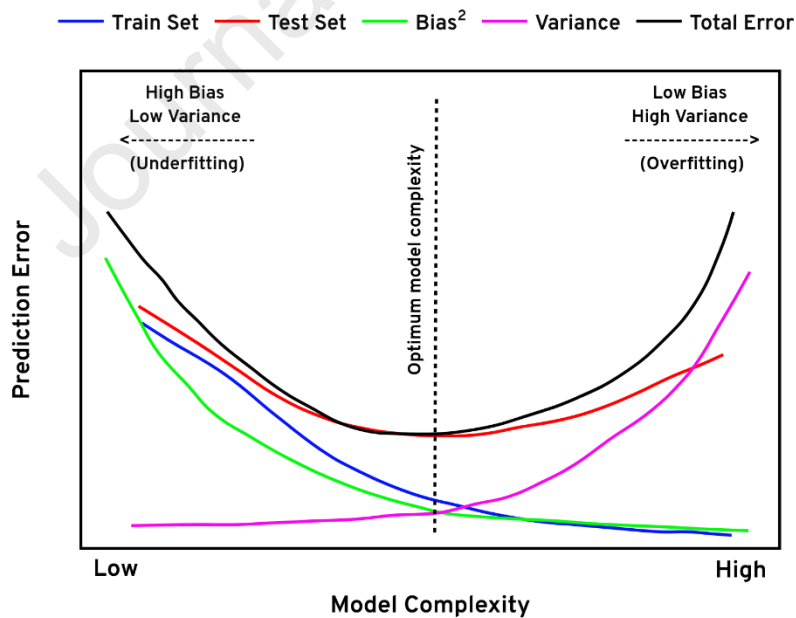


Figure 1. Bias-variance dilemma (Hastie et al., 2009; Fortmann-Roe, 2012)

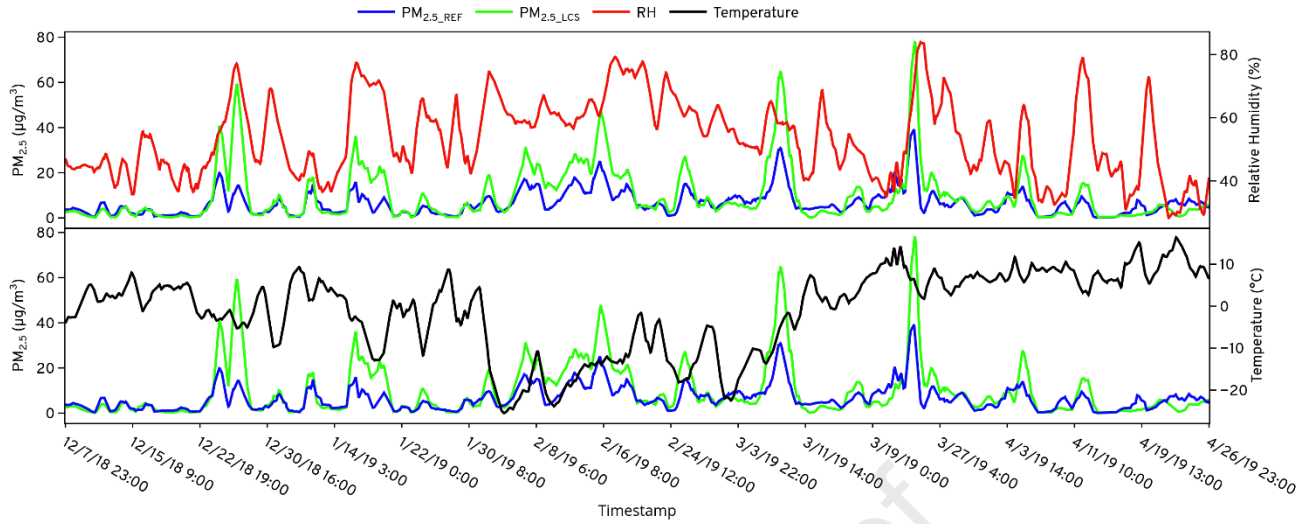


Figure 2. Effect of relative humidity and temperature on LCS $PM_{2.5}$ measurements

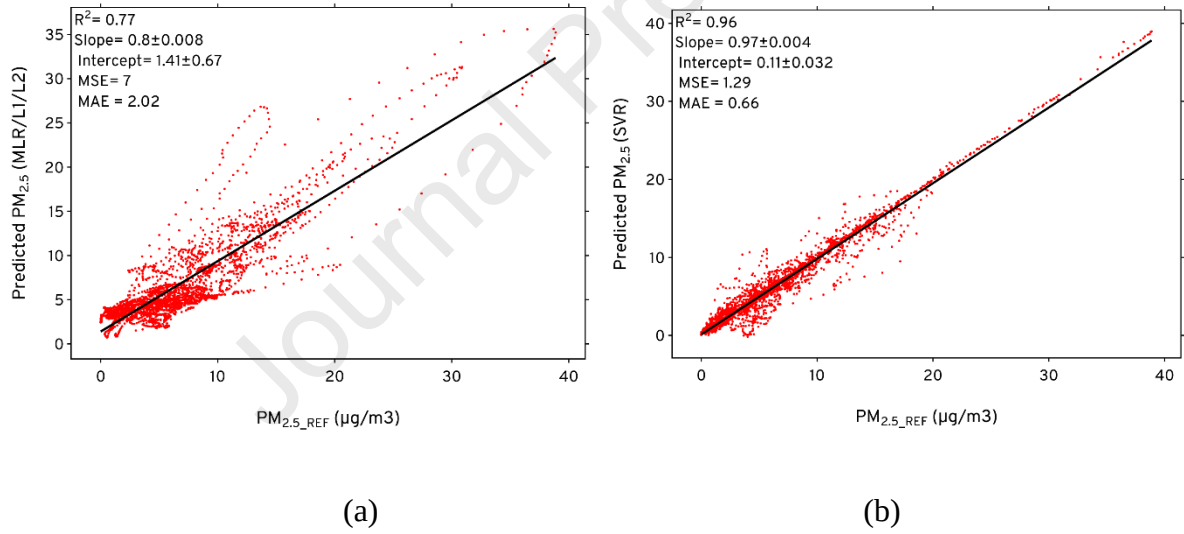


Figure 3. Scatter plot between $PM_{2.5_REF}$ and values predicted from (a) MLR/L1/L2, (b) SVR

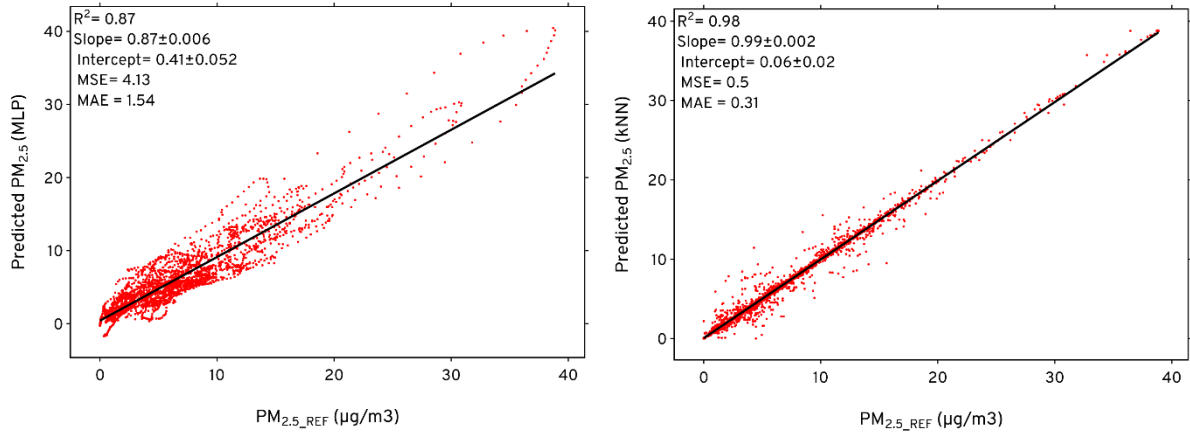
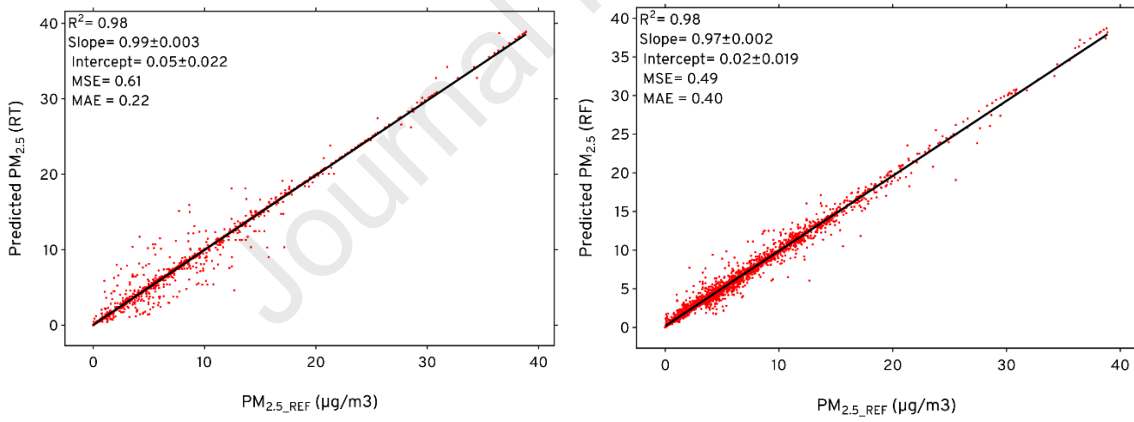
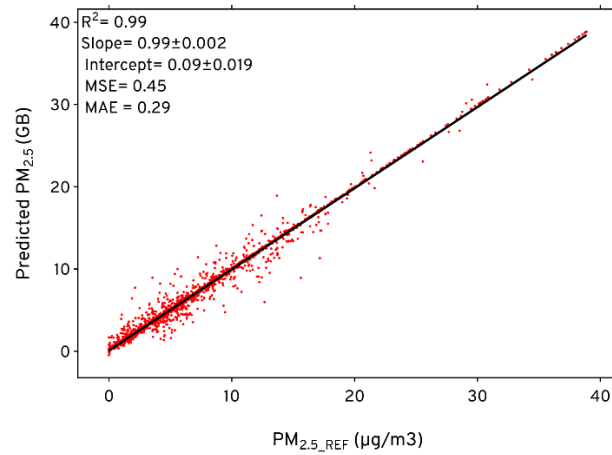
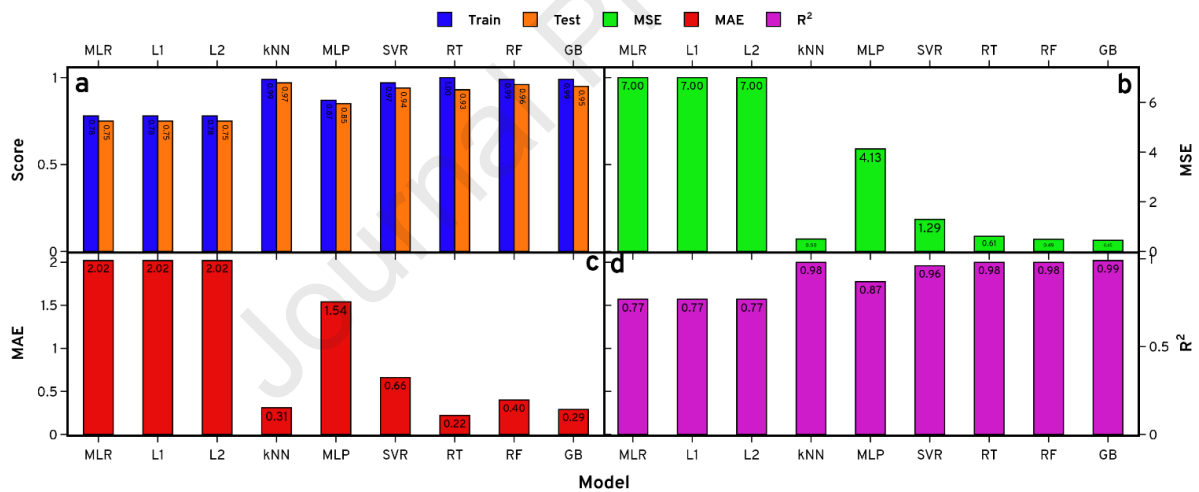


Figure 4. Scatter plot between $PM_{2.5_REF}$ and values predicted from (a) MLP, (b) kNN





(c)

Figure 5. Scatter plot between $PM_{2.5_REF}$ and values predicted from (a) RT, (b) RF, (c) GBFigure 6. Comparison of (a) train and test scores, (b) MSE, (c) MAE, (d) R^2 for the nine ML regression algorithms

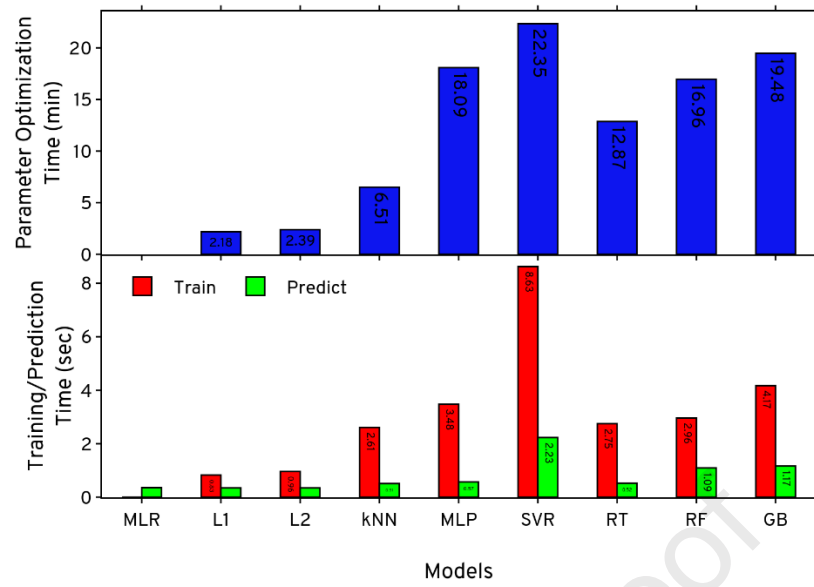


Figure 7. Comparison of computation time for the algorithms

Highlights:

- Evaluated nine regression algorithms for low-cost sensor calibration.
- Linear models are underfitting while regression tree is overfitting.
- k- Nearest Neighbour and Random Forest are providing accurate results.
- Gradient Boosting is also suited for low-cost sensor calibration.
- Validates the capability of machine learning algorithms for calibration.

Conflicts of Interest Statement

Manuscript title:

Evaluation of Nine Machine Learning Regression Algorithms for Calibration of Low-Cost
PM_{2.5} Sensor

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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